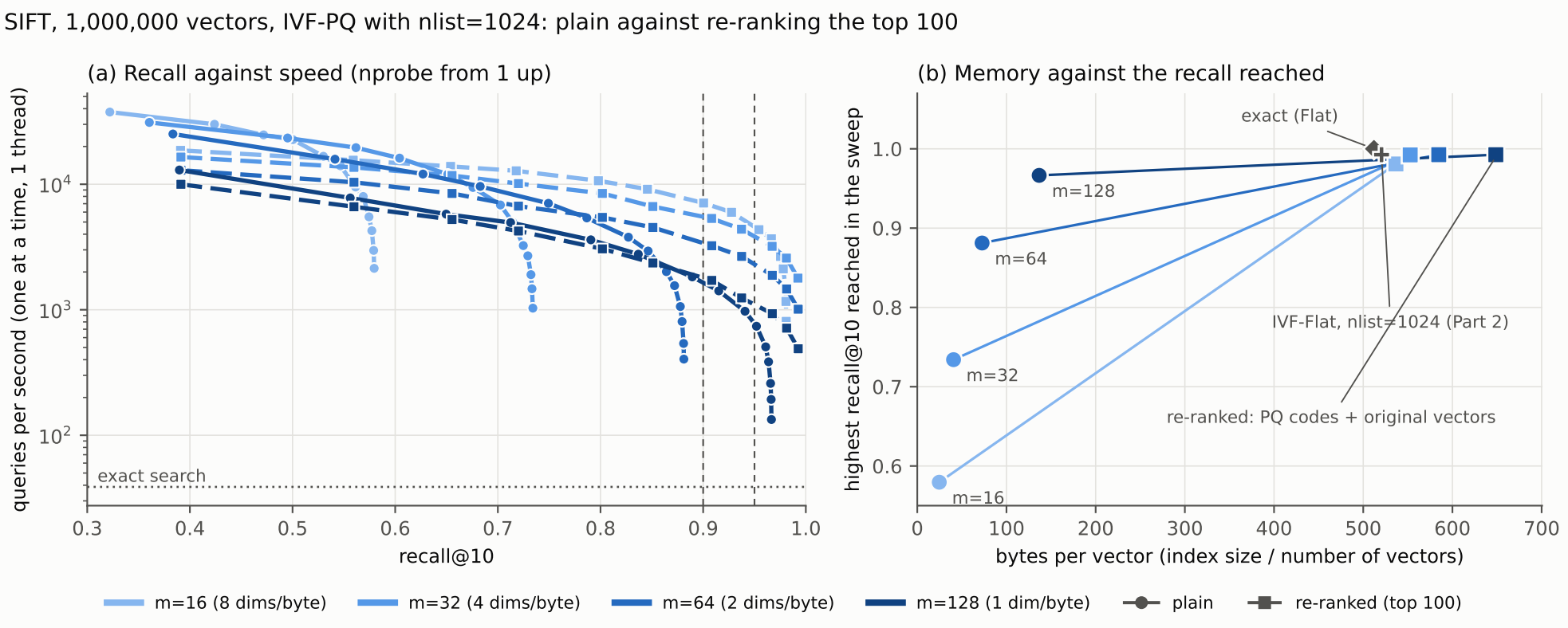
SIFT, 1,000,000 vectors, IVF-PQ with nlist=1024: plain against re-ranking the top 100
(a) Recall against speed (nprobe from 1 up)
queries per second (one at a time, 1 thread)
10^4
10^3
10^2
exact search
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
recall@10
(b) Memory against the recall reached
highest recall@10 reached in the sweep
1.0
0.9
0.8
0.7
0.6
exact (Flat)
m=128
m=64
m=32
m=16
IVF-Flat, nlist=1024 (Part 2)
re-ranked: PQ codes + original vectors
0
100
200
300
400
500
600
700
bytes per vector (index size / number of vectors)
m=16 (8 dims/byte)
m=32 (4 dims/byte)
m=64 (2 dims/byte)
m=128 (1 dim/byte)
plain
re-ranked (top 100)